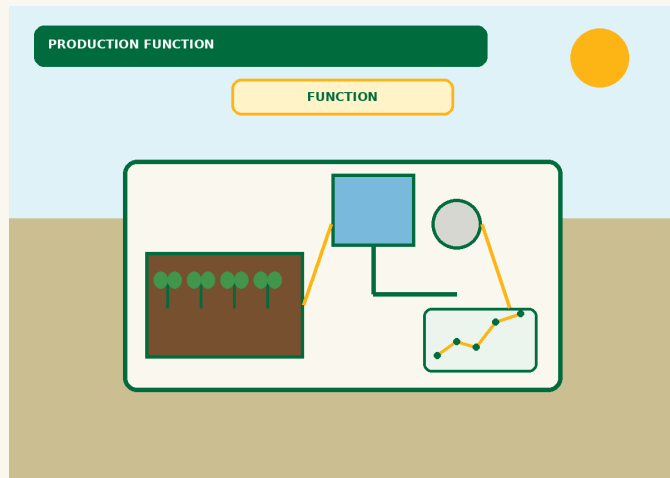




FUNCTION

PRODUCTION FUNCTION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

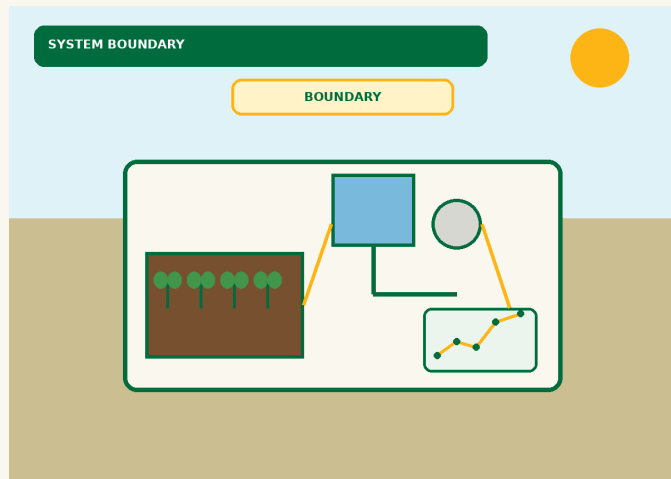
TECHNICAL CONCEPT

A production function represents the relationship between one or more inputs and resulting agricultural output.

ANALYSIS / DESIGN TASK

Identify inputs and outputs that could form a garden production function.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



BOUNDARY

SYSTEM BOUNDARY

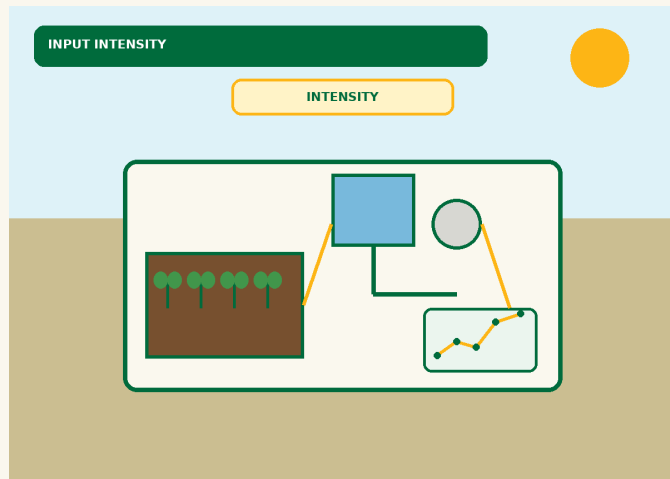
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A production-system boundary defines which material, resource, biological, and management flows are included in analysis.

ANALYSIS / DESIGN TASK

Draw and justify a boundary for one crop-production unit.



INTENSITY

INPUT INTENSITY

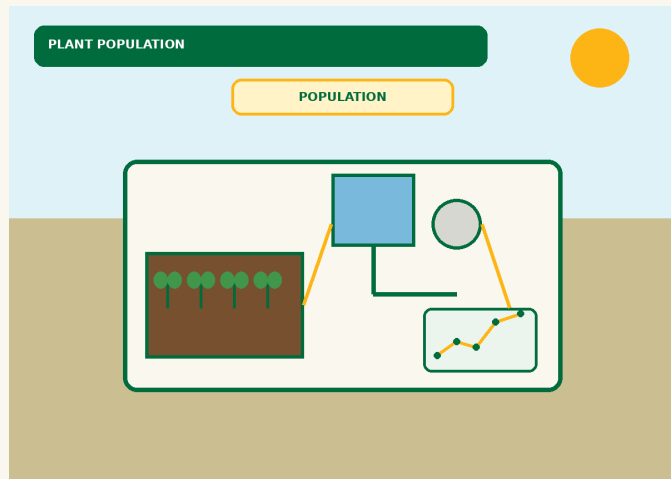
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Input intensity expresses the quantity of an input used per unit area, time, plant population, or output.

ANALYSIS / DESIGN TASK

Calculate water or nutrient input intensity.



POPULATION

PLANT POPULATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

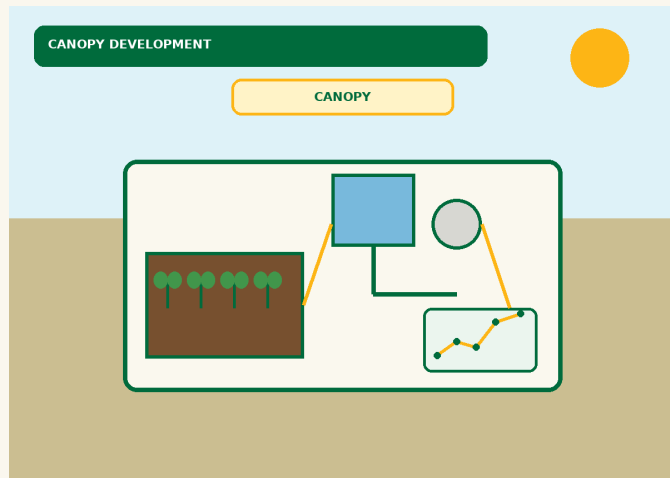
TECHNICAL CONCEPT

Plant population is determined by establishment, spacing, mortality, and replacement within a defined production area.

ANALYSIS / DESIGN TASK

Calculate actual and target plant populations.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



CANOPY

CANOPY DEVELOPMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

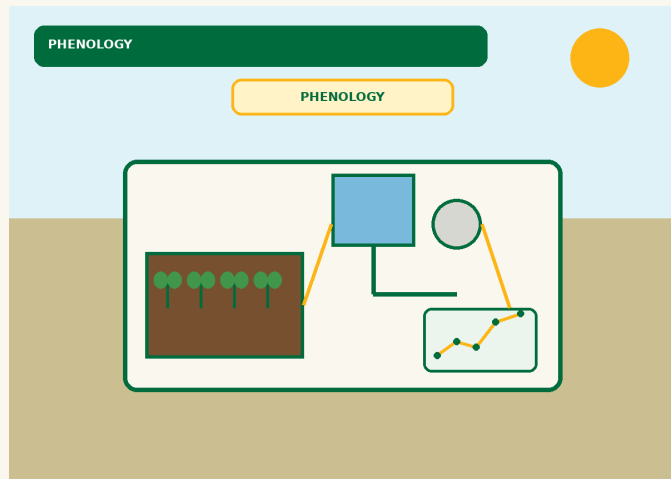
TECHNICAL CONCEPT

Canopy development describes changes in leaf area, architecture, and ground cover that influence resource capture.

ANALYSIS / DESIGN TASK

Track a canopy indicator through time.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



PHENOLOGY

PHENOLOGY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

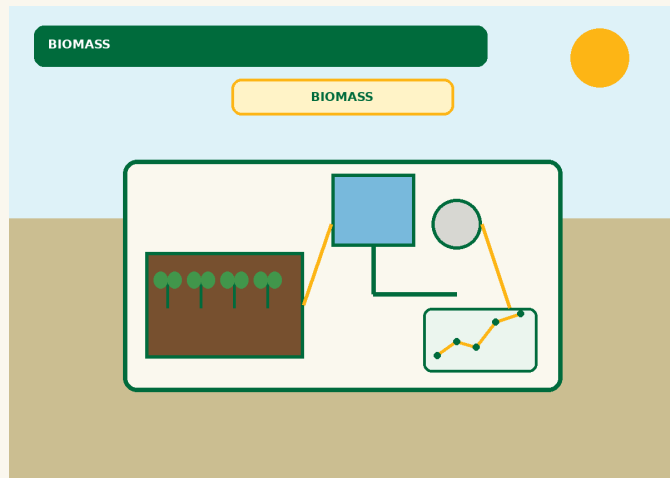
TECHNICAL CONCEPT

Phenology is the timing and progression of biological development stages such as emergence, flowering, fruiting, and maturity.

ANALYSIS / DESIGN TASK

Construct a crop phenology timeline.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



BIOMASS

BIOMASS

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

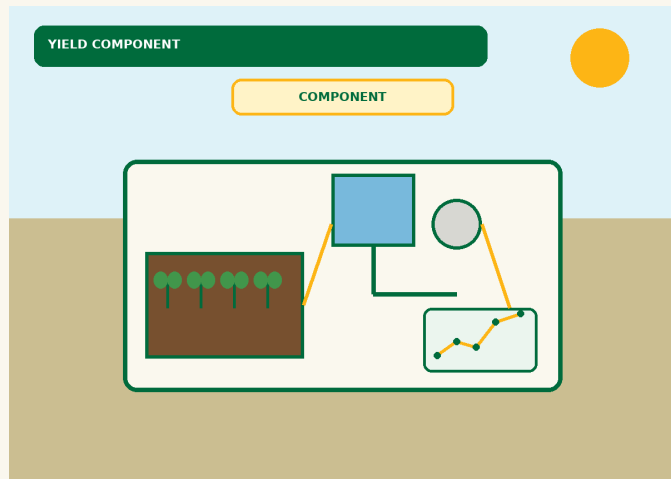
TECHNICAL CONCEPT

Biomass is the mass of living plant material and can be partitioned among roots, stems, leaves, and harvested organs.

ANALYSIS / DESIGN TASK

Interpret teacher-provided biomass data.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



COMPONENT

YIELD COMPONENT

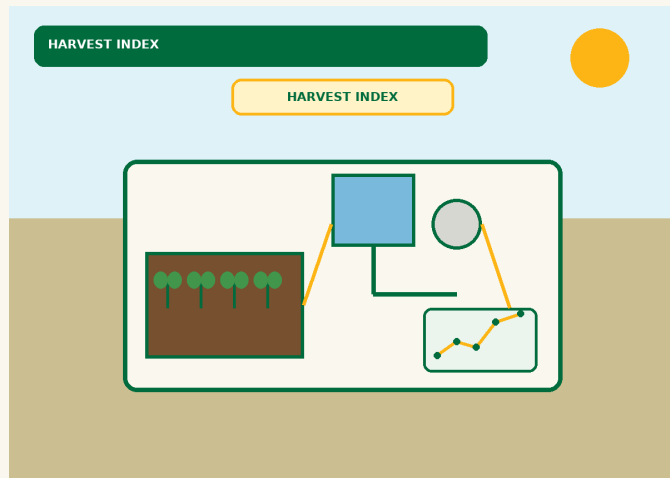
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Yield can be decomposed into components such as plant population, fruits per plant, and average fruit mass.

ANALYSIS / DESIGN TASK

Calculate yield from component data.



HARVEST INDEX

HARVEST INDEX

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

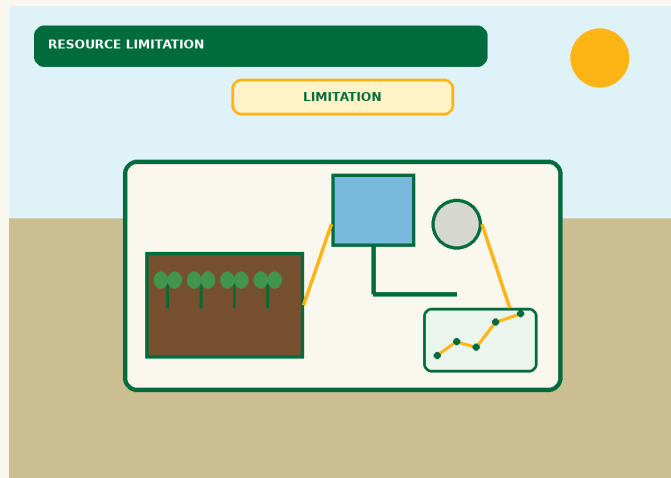
TECHNICAL CONCEPT

Harvest index is the ratio of economically useful harvested biomass to total aboveground biomass under a defined basis.

ANALYSIS / DESIGN TASK

Calculate harvest index from sample data.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



LIMITATION

RESOURCE LIMITATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

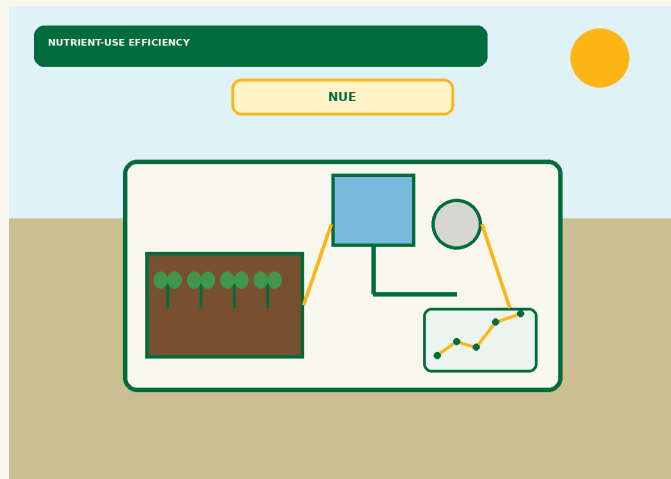
TECHNICAL CONCEPT

A limiting resource or condition constrains production when its availability or suitability is insufficient relative to crop demand.

ANALYSIS / DESIGN TASK

Diagnose a likely limiting factor from measurements.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



NUE

NUTRIENT-USE EFFICIENCY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Nutrient-use efficiency relates crop output or nutrient recovery to nutrient input using a clearly specified metric.

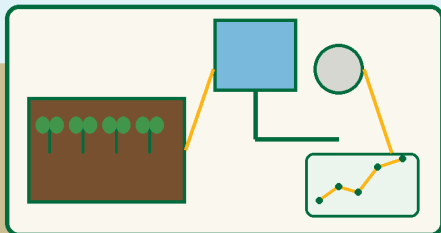
ANALYSIS / DESIGN TASK

Calculate a simplified nutrient-use metric.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

WATER PRODUCTIVITY

WP



WP

WATER PRODUCTIVITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

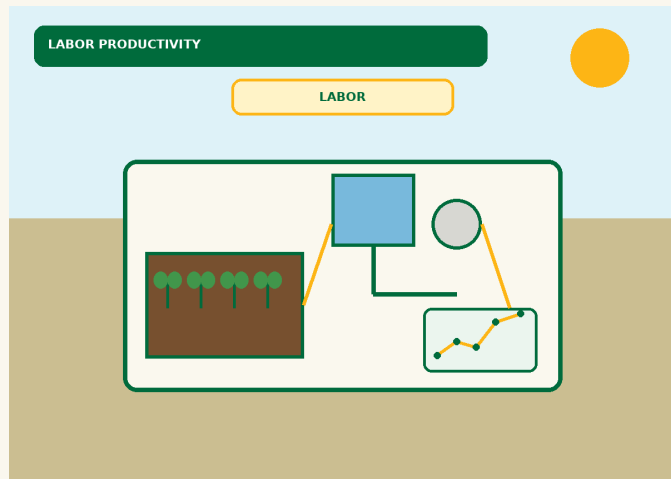
TECHNICAL CONCEPT

Water productivity relates useful crop output to water consumed or supplied, depending on the stated denominator.

ANALYSIS / DESIGN TASK

Calculate and compare water productivity.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



LABOR

LABOR PRODUCTIVITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

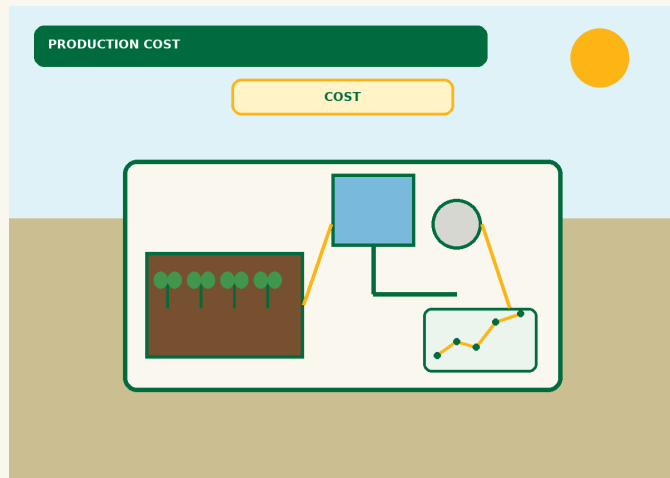
TECHNICAL CONCEPT

Labor productivity relates useful output to labor time or effort used in production.

ANALYSIS / DESIGN TASK

Calculate harvest per labor-hour.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



COST

PRODUCTION COST

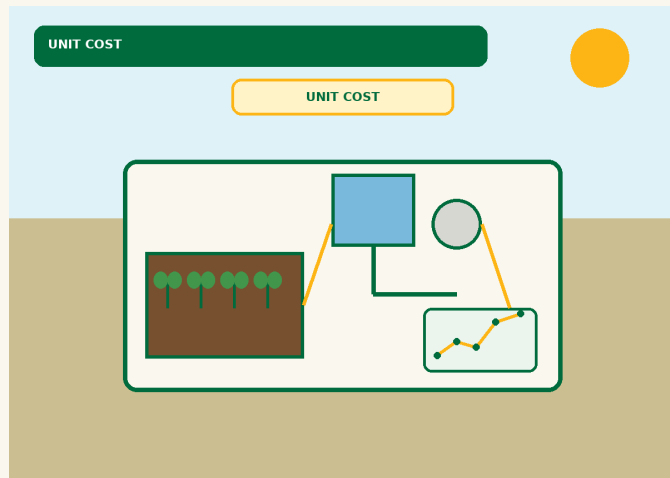
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Production cost includes variable and fixed resource expenditures attributable to a production cycle.

ANALYSIS / DESIGN TASK

Classify and total simplified cost data.



UNIT COST

UNIT COST

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

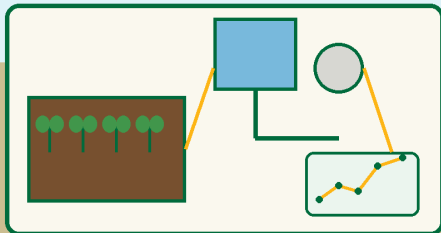
Unit cost is the production cost divided by the quantity of useful output produced.

ANALYSIS / DESIGN TASK

Calculate cost per kilogram of harvest.

LOSS ANALYSIS

LOSSES



LOSSES

LOSS ANALYSIS

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

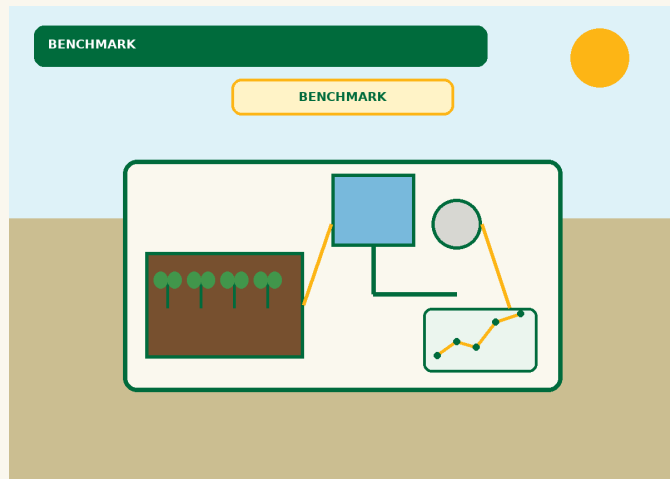
TECHNICAL CONCEPT

Loss analysis identifies where expected production is reduced during establishment, growth, harvest, handling, or storage.

ANALYSIS / DESIGN TASK

Quantify one loss pathway.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



BENCHMARK

BENCHMARK

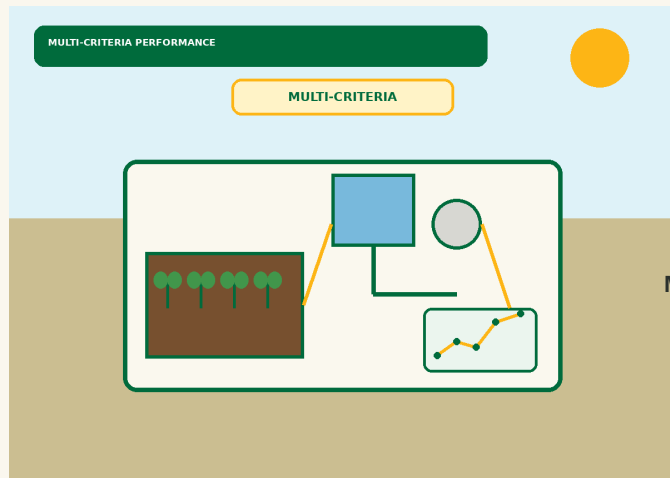
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A benchmark is a reference performance value used to compare current results with a target, prior cycle, or alternative system.

ANALYSIS / DESIGN TASK

Compare current performance with a stated benchmark.



MULTI-CRITERIA

MULTI-CRITERIA PERFORMANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Agricultural performance often requires simultaneous consideration of yield, quality, water, nutrients, cost, labor, and reliability.

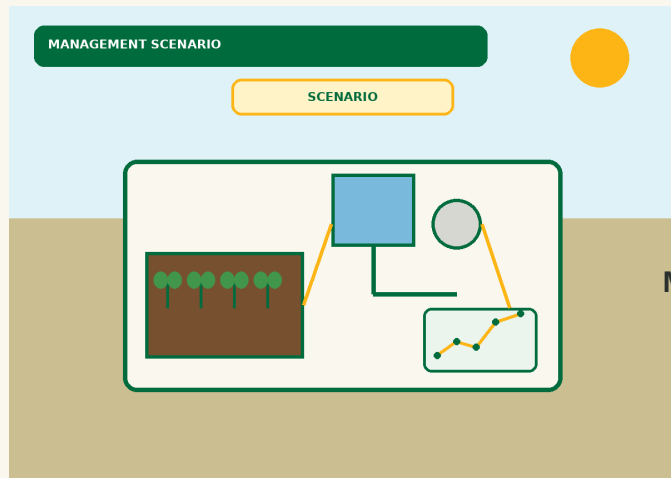
ANALYSIS / DESIGN TASK

Construct a simple performance scorecard.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

SCENARIO

MANAGEMENT SCENARIO



DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

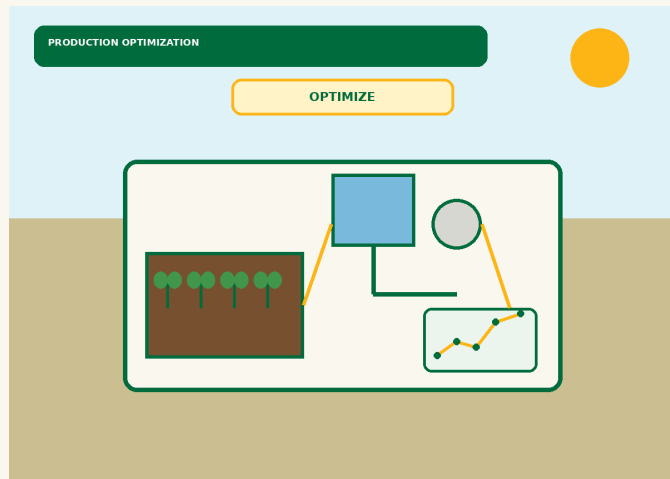
TECHNICAL CONCEPT

A management scenario specifies a plausible combination of inputs, practices, constraints, and operating rules for comparison.

ANALYSIS / DESIGN TASK

Compare two production scenarios.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



OPTIMIZE

PRODUCTION OPTIMIZATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Production optimization seeks a feasible management combination that improves defined objectives while respecting constraints and trade-offs.

ANALYSIS / DESIGN TASK

Recommend an optimized strategy and state its trade-offs.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE